3. GATE-FIRST STRUCTURE AND PROCESS EDITOR IMAGES

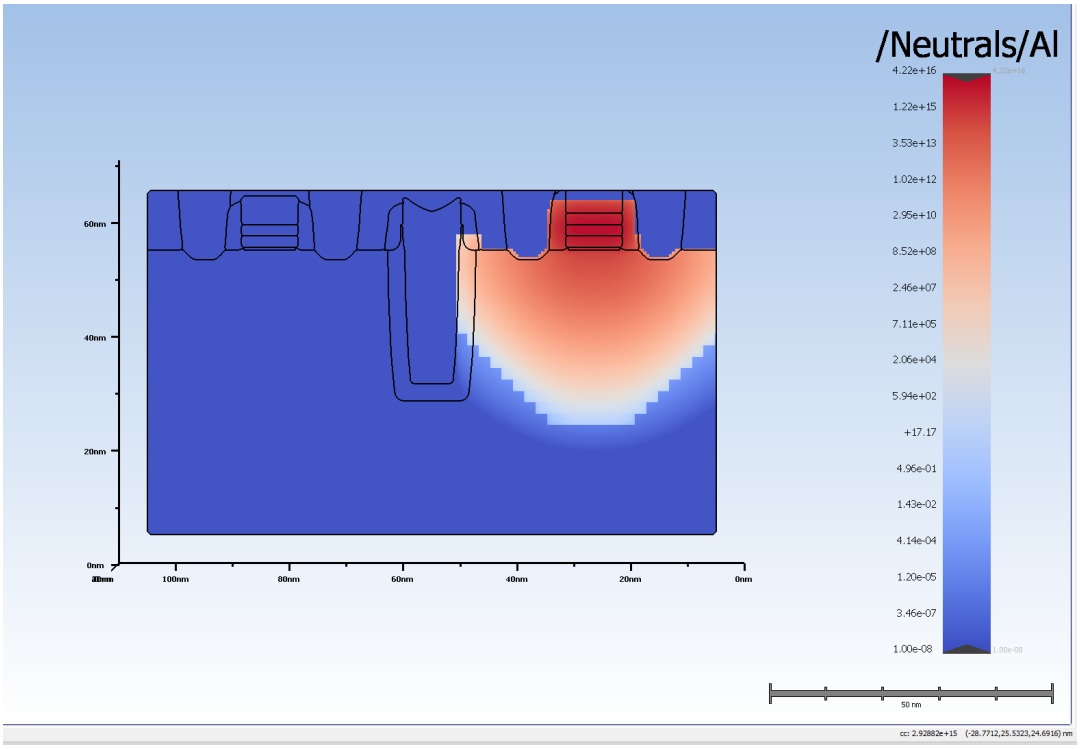


Figure 1 : Al Metal Diffusion in Gate-First Structure

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	/Silicon/Si_Xtal		50			Wafer1				
2		Conformal ...	Pad Oxide Depo...	/Oxides/SiO2_T...		5			Wafer1	Top	1.0		
3		Conformal ...	SiN hardmask	/Nitrides/Si3N4...		10			Wafer1	Top	1.0		
4		Sequence	STI_Trench_Lith...										
4.1		Planarizing ...	Photo Resist co...	/Organics/Resist		40			Wafer1	Top			
4.2		Expose Mat...	Expose STI_Tren...	/Organics/Resist		0	STI_Trench	Dark	Wafer1	Top			
5		Sequence	Etching trench										
5.1		Enhanced B...	SiN hardmask e...	/Nitrides/, /Org...		15			Wafer1	Top	0		
5.2		Enhanced B...	pad oxide Etch	/Nitrides/, /Org...		5			Wafer1	Top	0		
5.3		Enhanced B...	substrate silico...	/Nitrides/, /Org...		25			Wafer1	Top	0.05		
6		Remove M...	resist removal	/Organics/, /Or...					Wafer1				
7		Interface Gr...	Thermal Oxidati...	/Air, /Silicon/					Wafer1				
8		Conformal ...	Filling trench wi...	/Oxides/SiO2_P...		10			Wafer1	Top	1		
9		Polish	STI oxide CMP						Wafer1	Top			
10		Enhanced B...	Removal of SiN ...	/Nitrides/Si3N4...		10			Wafer1	Top	0		
11		Multi Etch (...	Pad_oxide ramo...	/Oxides/SiO2_P...		5			Wafer1	Top			Unknown Step ...
12		Conformal ...	Gate oxide dep...	/Oxides/SiO2_H...		0.5			Wafer1	Top	1.0		
13		Conformal ...	High k depositi...	/Oxides/HfOx		2			Wafer1	Top	1.0		
14		Conformal ...	NMOS gate metal	/Metals/TiAl	/Neutrals/Al	2			Wafer1	Top	1.0		
15		Sequence	gate patterning										
15.1		Sequence	NMOS Gate_Lit...										
15.1.1		Planarizing ...	Photo Resist co...	/Organics/Resist		5			Wafer1	Top			
15.1.2		Expose Mat...	Expose for NM...	/Organics/Resist		0	NMOS_metal_p...	Dark	Wafer1	Top			
15.2		Enhanced B...	NMOS Metal Re...	/Metals/TiAl, /...		2			Wafer1	Top	1		
15.3		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
16		Conformal ...	gate metal dep...	/Metals/TiN		2			Wafer1	Top	1.0		
17		Sequence	NMOS PMOS g...										
17.1		Conformal ...	Final Gate meta...	/Metals/W		5			Wafer1	Top	1.0		
17.2		Planarizing ...	patterning NM...	/Organics/Resist		10			Wafer1	Top			
17.3		Expose Mat...	Expose for NM...	/Organics/Resist		0	Gate_both_NM...	Light	Wafer1	Top			
17.4		Enhanced B...	W Etch	/Metals/W, /Or...		5			Wafer1	Top	0.0		
17.5		Enhanced B...	TiN/TiAl Etch	/Metals/TiAl, /...		5			Wafer1	Top	0.01		
17.6		Enhanced B...	high k Etch	/Metals/TiAl, /...		2			Wafer1	Top	0.0		
17.7		Enhanced B...	Gate oxide Etch	/Metals/TiAl, /...		2			Wafer1	Top	0.0		
17.8		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
18		Sequence	NMOS source a...										
18.1		Conformal ...	Conformal Dep...	/Nitrides/Si3N4...		3			Wafer1	Top	1.0		
18.2		Enhanced B...	Basic Etch	/Nitrides/		4			Wafer1	Top	0.001		
18.3		Planarizing ...	Resist pattern f...	/Organics/Resist		10			Wafer1	Top			
18.4		Expose Mat...	Expose resist for...	/Organics/Resist		0	Source_Drain_N...	Dark	Wafer1	Top			
18.5		Implant (A...	NMOS source a...		/NType/P				Wafer1	Top			
18.6		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
19		Sequence	PMOS source a...										
19.1		Planarizing ...	Resist pattern f...	/Organics/Resist		10			Wafer1	Top			
19.2		Expose Mat...	Expose resist for...	/Organics/Resist		0	Source_Drain_P...	Dark	Wafer1	Top			
19.3		Implant (A...	PMOS source a...		/PType/B				Wafer1	Top			
19.4		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
19.5		Diffusion (...	Diffusion (Requ...	/Oxides/HfOx, /...	/NType/P, /Neu...				Wafer1				12
20		Sequence	Source drain m...										
20.1		Planarizing ...	interlayer dielec...	/Oxides/SiO2_LTO		10			Wafer1	Top			
20.2		Planarizing ...	ressit pattern fo...	/Organics/Resist		10			Wafer1	Top			
20.3		Expose Mat...	Expose resist for...	/Organics/Resist		0	metal_NMOS_P...	Dark	Wafer1	Top			
20.4		Enhanced B...	source and drai...	/Nitrides/, /Org...		30			Wafer1	Top	0.1		
20.5		Remove M...	resist removal a...	/Organics/, /Or...					Wafer1				
20.6		Conformal ...	TiN Deposit	/Metals/TiN		12			Wafer1	Top	1.0		
21		Polish	Final CMP of ILD						Wafer1	Top			

Figure 2 : Process Editor Steps for Gate-First Flow

3. ANNEAL TIME VS. AL CONTAMINATION PROFILE

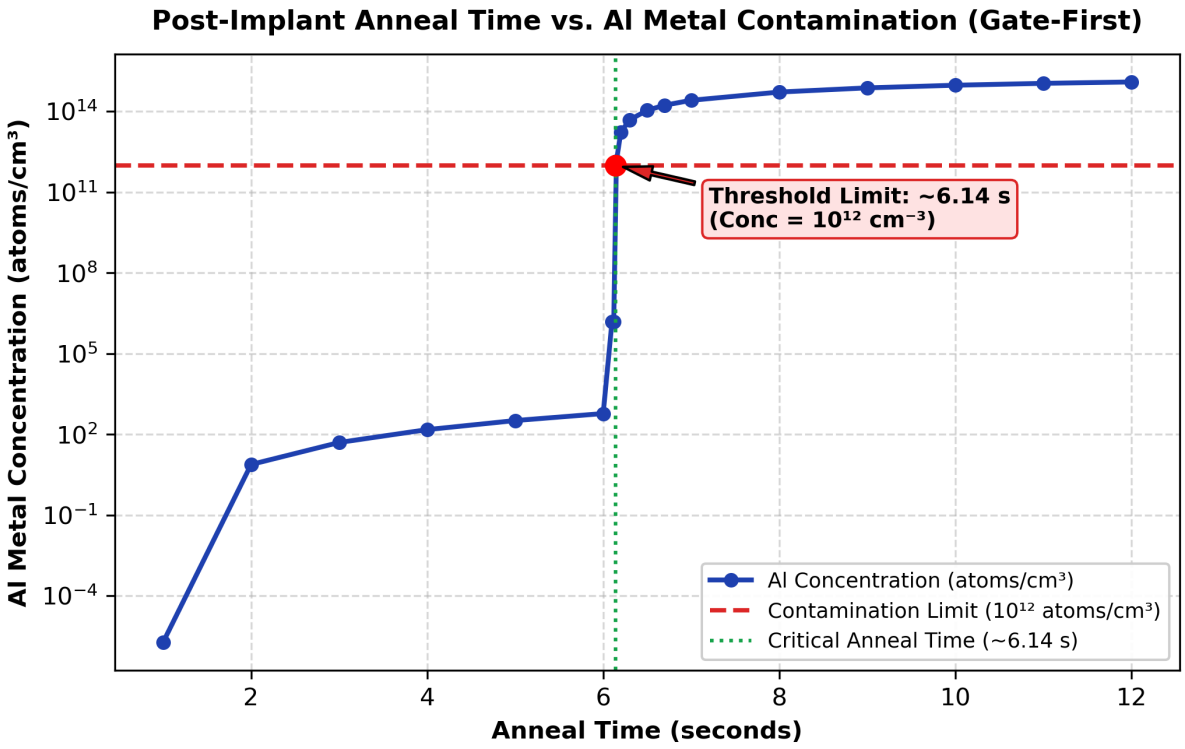


Figure 3 : Post-Implant Anneal Time vs. Al Metal Contamination

3. CROSS-SECTIONAL AL DIFFUSION PROFILES AT VARYING ANNEAL TIMES

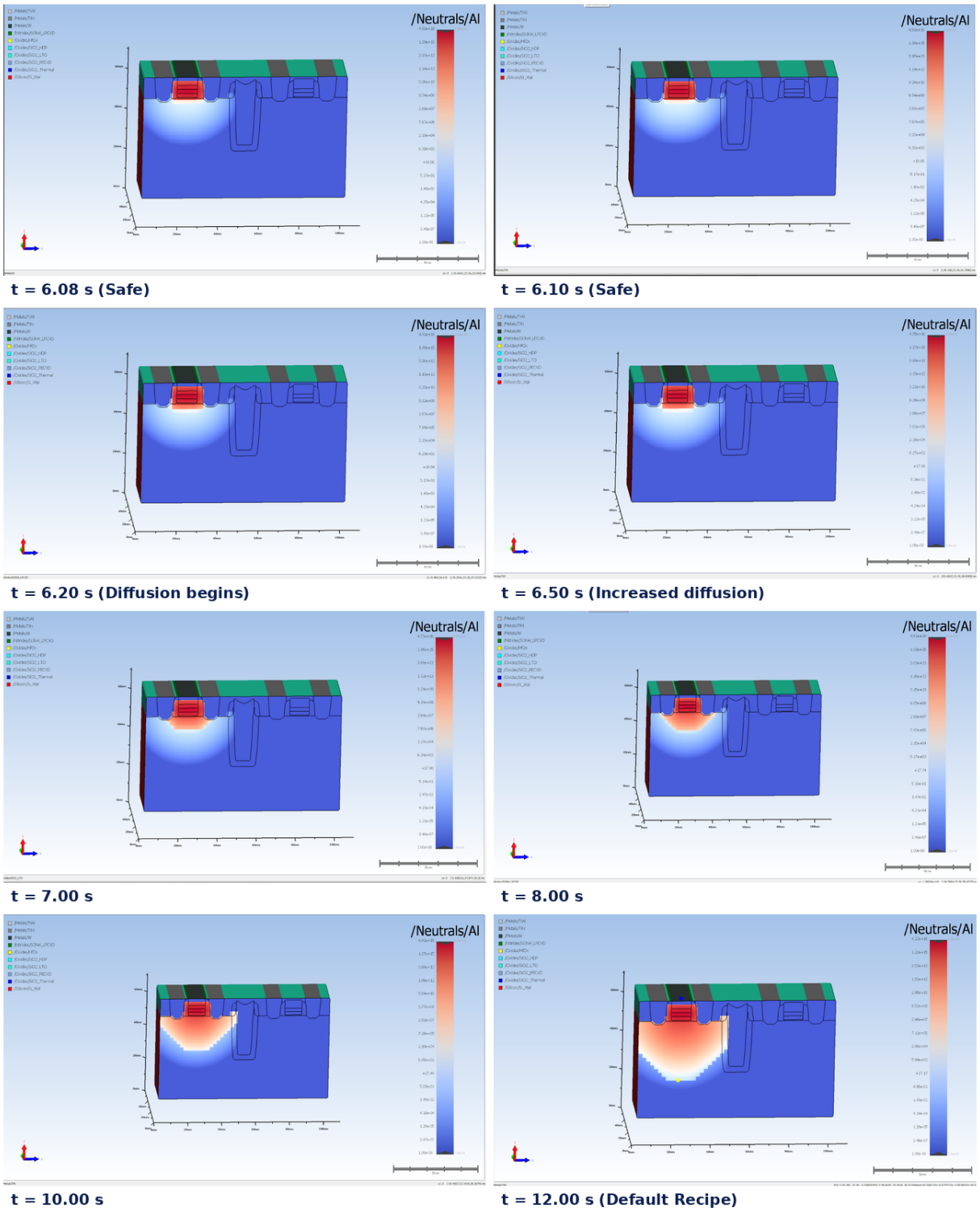


Figure 4 : Cross-Sectional Al Diffusion Profiles at Varying Anneal Times (6.08 s to 12.0 s)

3. GATE OXIDE THICKNESS VS. AL CONTAMINATION PROFILE

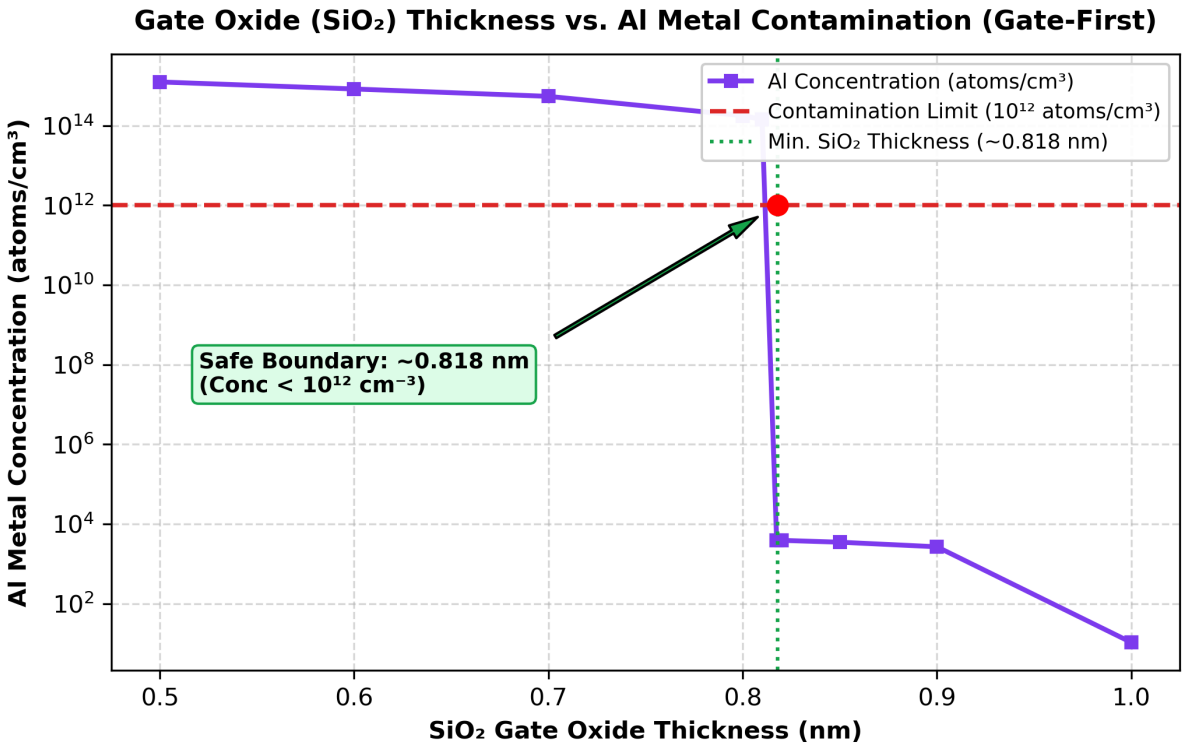


Figure 5 : Gate Oxide Thickness vs. Al Metal Contamination

3. CROSS-SECTIONAL AL DIFFUSION PROFILES AT VARYING GATE OXIDE THICKNESSES

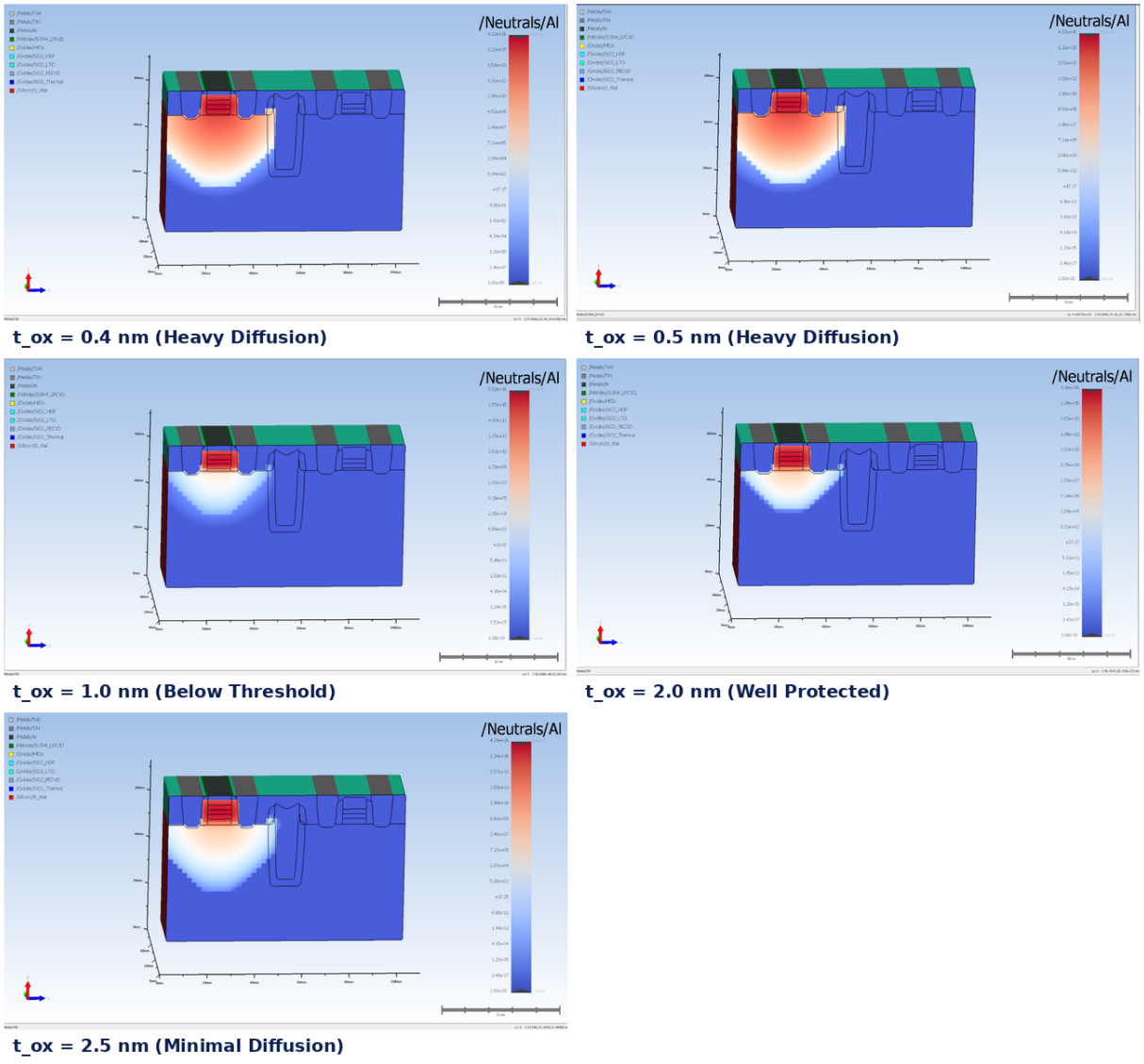


Figure 6 : Cross-Sectional Al Diffusion Profiles at Varying Gate Oxide Thicknesses (0.4 nm to 2.5 nm)

4. GATE-LAST STRUCTURE AND PROCESS EDITOR IMAGES

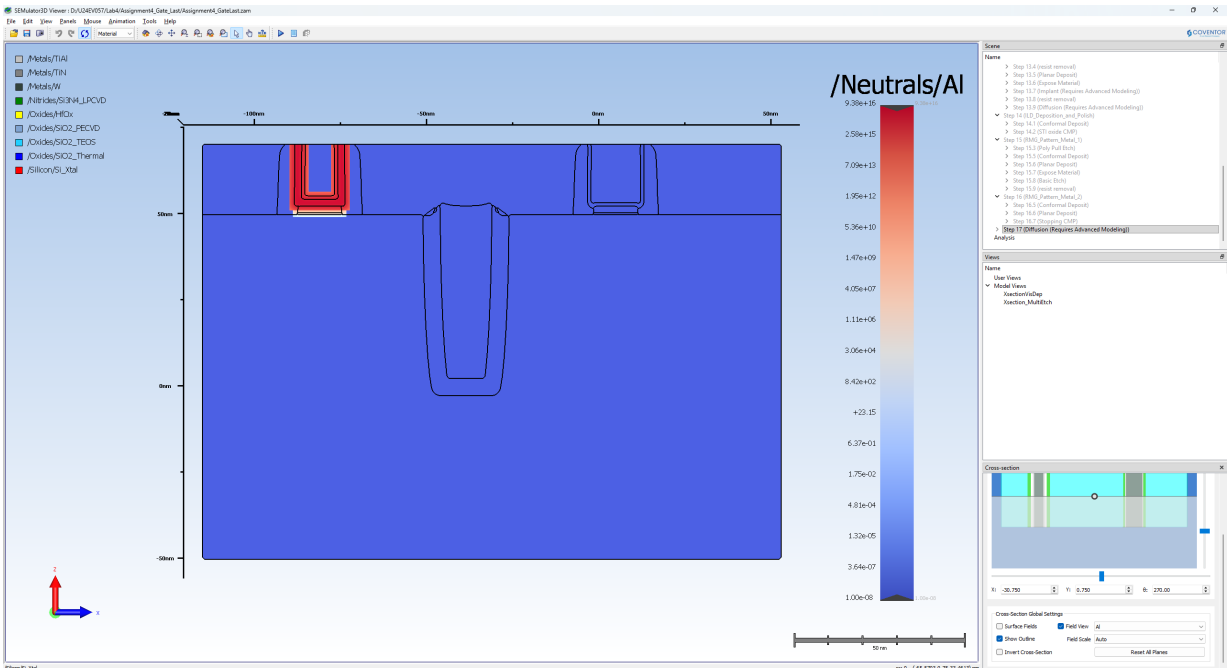


Figure 7 : Al Metal Diffusion in Gate-Last Structure

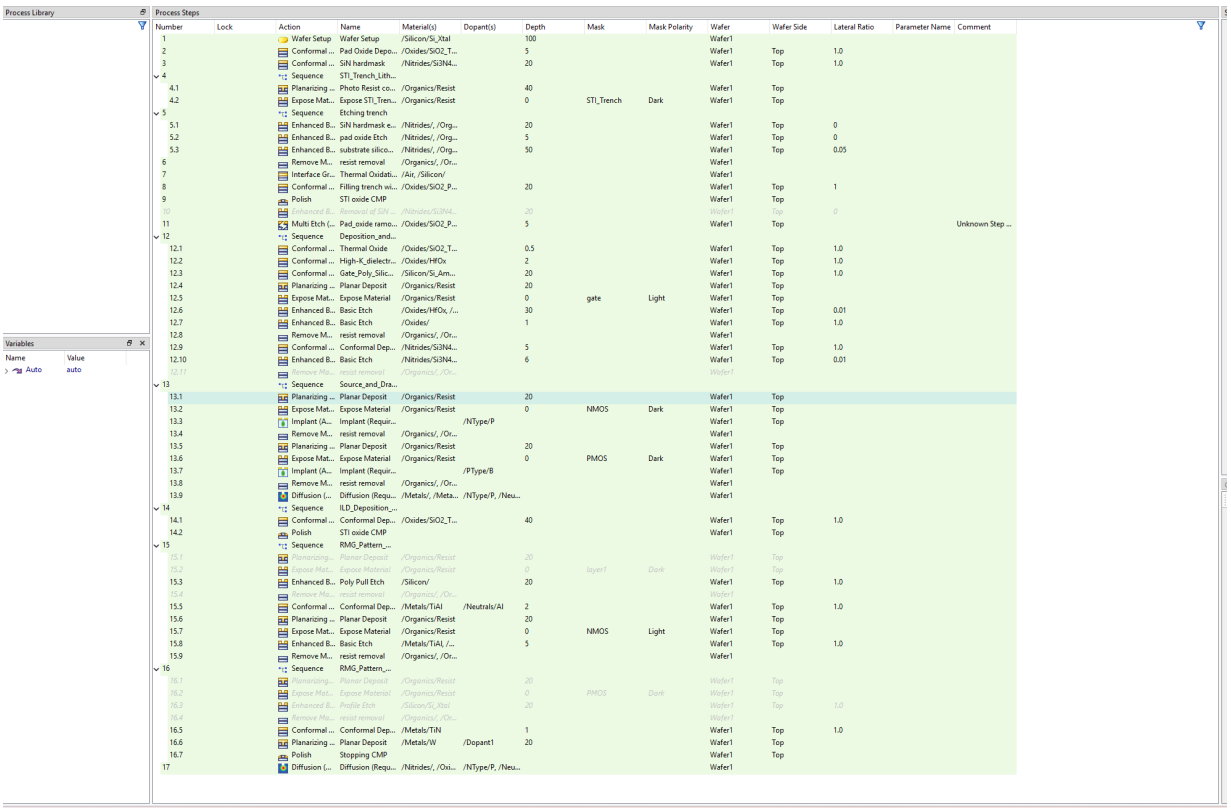


Figure 8 : Process Editor Steps for Gate-Last Flow

5. 14 NM FINFET SACRIFICIAL GATE & POST-RTA STRUCTURE IMAGES

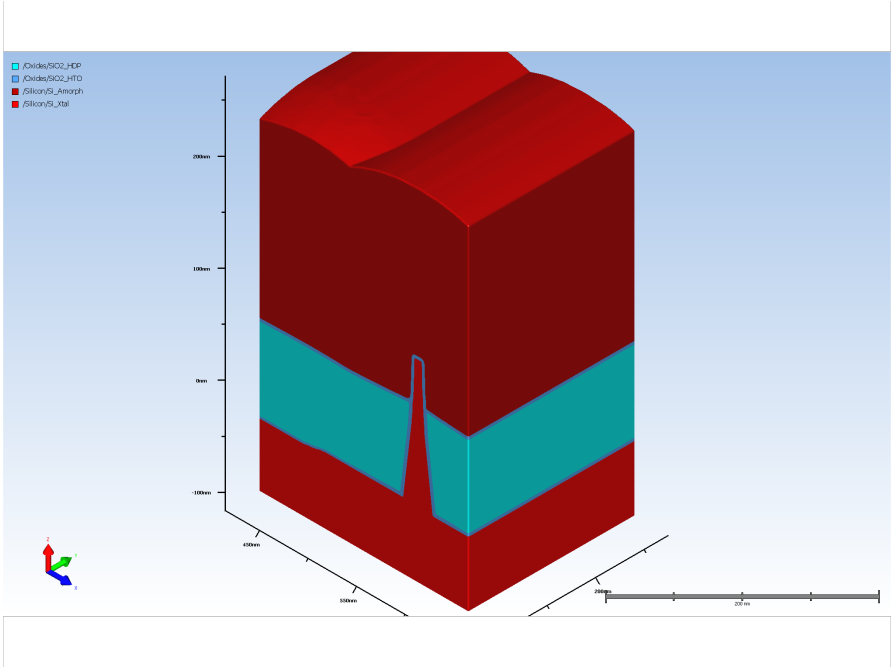


Figure 9 : Sacrificial Gate Deposition in 14 nm FinFET (Step 2.5.3)

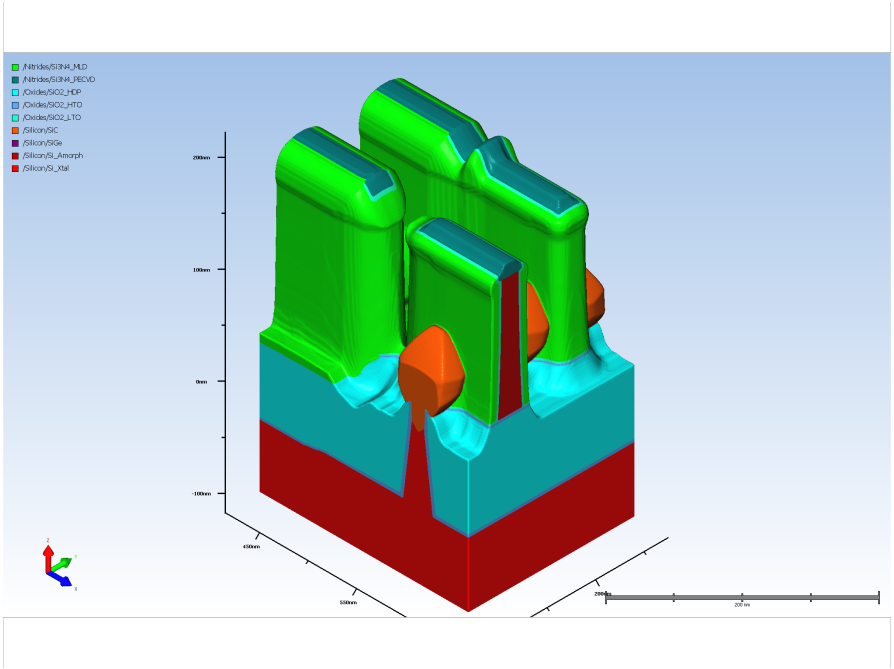


Figure 10 : 14 nm FinFET Structure Post-RTA Prior to RMG (Step 2.6.3)